

SE2030 - Software Engineering

Week 2 Quiz: Software Development Models (SDLC vs. Process Models)
60 Questions | Mix of Multiple Choice (MCQ) and Multiple Select (MSQ)

Instructions: Each question is marked as **[MCQ]** (choose ONE correct answer) or **[Select all that apply]** (choose ALL correct answers). Questions cover the lecture slides and the lecture recording (Waterfall, Iterative Waterfall, V-Model, Prototyping, Incremental, Spiral, and Agile models). The answer key is provided at the end.

1. What does SDLC (Software Development Life Cycle) primarily define?

[MCQ - choose one]

- A) How to execute the phases of software development
- B) What phases to follow when developing software
- C) Which programming language to use
- D) The pricing model for the software

2. What does a Software Process Model primarily define?

[MCQ - choose one]

- A) What phases to follow
- B) How to execute the SDLC phases
- C) The legal requirements of the project
- D) The stakeholders involved

3. Which of the following are given as examples of Software Process Models in the lecture?

[Select all that apply]

- A) Waterfall
- B) Spiral
- C) Agile
- D) Feasibility Study

4. According to the 'How Are They Connected?' comparison, SDLC is described as:

[MCQ - choose one]

- A) A specific implementation
- B) A general structure
- C) A type of testing
- D) A programming framework

5. In the house-building analogy from the lecture recording, what do the SDLC phases represent?

[MCQ - choose one]

- A) The specific brand of materials used
- B) The basic steps that never change, regardless of how the house is built
- C) The architect's personal preference
- D) The cost of the house

6. In the house-building analogy, what does the 'software process model' represent?

[MCQ - choose one]

- A) The basic steps of building (foundation, walls, roof)
- B) The way those basic steps are carried out (all at once vs. floor by floor)
- C) The stakeholders of the house

D) The feasibility study of the house

7. Which software process model executes SDLC sequentially, according to the lecture?

[MCQ - choose one]

- A) Agile model
- B) Waterfall model
- C) Spiral model
- D) Incremental model

8. Which software process model executes SDLC in iterations (sprints)?

[MCQ - choose one]

- A) Waterfall model
- B) V-Model
- C) Agile model
- D) Classical Waterfall model

9. According to the 'Evolution of Software Process Models' table, which descriptions are correctly matched to their model?

[Select all that apply]

- A) Waterfall Model - Linear and phase-based
- B) V-Model - Emphasizes validation and testing at each phase
- C) Incremental Model - Software delivered in small parts or versions
- D) Spiral Model - Delivered in small parts only, with no risk assessment

10. In the Waterfall Model, what is the correct order of phases?

[MCQ - choose one]

- A) Design -> Requirements -> Implementation -> Testing -> Deployment -> Maintenance
- B) Requirements -> Design -> Implementation -> Testing -> Deployment -> Maintenance
- C) Testing -> Requirements -> Design -> Implementation -> Deployment -> Maintenance
- D) Requirements -> Implementation -> Design -> Testing -> Deployment -> Maintenance

11. Which of the following are true about the Classical Waterfall Model's handling of requirements, according to the lecture?

[Select all that apply]

- A) Requirements are frozen once gathered and approved
- B) There is no way to go back to a previous phase to fix an incorrect requirement
- C) Errors are typically only discovered in a later stage (Testing)
- D) Requirements can be updated at any time without restriction

12. According to the lecture recording, which version of the Waterfall Model is actually used in industry today?

[MCQ - choose one]

- A) Classical Waterfall Model only
- B) Neither version is used anymore
- C) Iterative Waterfall Model
- D) Both are used equally

13. What is a key limitation of the Waterfall Model mentioned in the lecture?

[MCQ - choose one]

- A) It is too flexible for simple projects

- B) It is not suitable for projects where requirements may change later
- C) It requires too much testing
- D) It cannot be used for any real project

14. According to the lecture recording, in the Classical Waterfall Model, when do requirement errors typically surface?

[MCQ - choose one]

- A) During the requirements phase itself
- B) During the design phase
- C) During the testing phase, which is too late to fix easily
- D) They never surface

15. Which system type is given as a good example for using the Waterfall Model?

[MCQ - choose one]

- A) A rapidly changing social media app
- B) Payroll or HR Management systems with stable, well-defined requirements
- C) A prototype-based mobile game
- D) A defense missile control system

16. What is the key improvement the Iterative Waterfall Model adds over the Classical Waterfall Model?

[MCQ - choose one]

- A) Removing the testing phase
- B) Allowing feedback and revisions (going back) between phases
- C) Combining design and coding into one phase
- D) Skipping the feasibility study

17. According to the lecture, the Iterative Waterfall Model is suitable when:

[Select all that apply]

- A) Requirements are mostly clear, but some minor changes may be needed
- B) Client feedback is available at the end of each phase
- C) The project is a medium-sized system with moderate risk
- D) The project requirements are completely unknown and constantly changing

18. According to the lecture recording, an experienced team building a common system (e.g., payroll) is well suited to which model?

[MCQ - choose one]

- A) Spiral Model
- B) Iterative Waterfall Model
- C) Prototyping Model
- D) Agile Model

19. What is the benefit of the 'feedback paths' (back arrows) in the Iterative Waterfall Model?

[MCQ - choose one]

- A) They let teams go back to earlier phases to fix problems found later
- B) They remove the need for testing
- C) They allow skipping the design phase
- D) They automatically fix bugs

20. For a Student Record Management System where requirements may change based on administrative staff feedback, which model does the lecture suggest?

[MCQ - choose one]

- A) Classical Waterfall Model
- B) Iterative Waterfall Model
- C) Spiral Model
- D) V-Model

21. Why is it called the 'V-Model'?

[MCQ - choose one]

- A) Because it stands for 'Very fast' development
- B) Because the diagram of the model forms a V shape
- C) Because it was invented by someone named V
- D) Because it only has version-based releases

22. What is the primary focus of the V-Model?

[MCQ - choose one]

- A) Rapid prototyping
- B) Testing and validation at every stage of development
- C) Risk analysis only
- D) Continuous customer feedback

23. In the V-Model diagram, what happens on the left side?

[MCQ - choose one]

- A) Testing and validation
- B) Planning and designing
- C) Deployment
- D) Maintenance

24. In the V-Model, Requirement Gathering corresponds to which testing type on the right side?

[MCQ - choose one]

- A) Unit Testing
- B) Integration Testing
- C) System Testing
- D) Acceptance Testing

25. In the V-Model diagram, which of the following are correct development-to-testing pairings?

[Select all that apply]

- A) Module Design <-> Unit Testing
- B) Software Design <-> Integration Testing
- C) System Analysis <-> System Testing
- D) Requirement Gathering <-> Coding

26. According to the lecture recording, when do QA engineers typically begin their work in a normal SDLC, compared to the V-Model?

[MCQ - choose one]

- A) Normally they start at the same time as developers in both cases
- B) Normally they start later (during testing), but in the V-Model they work in parallel with developers from the start

- C) In the V-Model, QA engineers work only after deployment
- D) QA engineers are not needed in the V-Model

27. In the V-Model, what do QA engineers create immediately after reviewing the SRS (Software Requirements Specification) document?

[MCQ - choose one]

- A) The final test report
- B) User acceptance test plans and test cases
- C) The source code
- D) The deployment plan

28. Which types of systems is the V-Model best suited for, according to the lecture?

[MCQ - choose one]

- A) Simple prototypes
- B) Safety-critical systems like medical devices and automotive systems
- C) Small social media apps
- D) Quick internal tools

29. What is the main purpose of the Prototyping Model?

[MCQ - choose one]

- A) To deliver the final product as quickly as possible without feedback
- B) To build a quick, rough version of the system to gather accurate feedback before building the final system
- C) To skip requirements gathering entirely
- D) To focus only on risk analysis

30. What are the steps in the Prototyping Model, according to the lecture?

[Select all that apply]

- A) Gather basic requirements
- B) Build a quick prototype
- C) Show it to the user and get feedback
- D) Immediately deploy the prototype as the final system without any changes

31. According to the lecture recording, what typically happens to the prototype once requirements are correctly understood?

[MCQ - choose one]

- A) It is delivered to the client as the final system
- B) It is usually discarded, and the real system is built from scratch
- C) It is sold separately as a product
- D) It replaces the SRS document

32. Why might a client struggle to clearly explain their requirements, according to the lecture recording?

[MCQ - choose one]

- A) They are intentionally hiding information
- B) They may not be technically sound or computer literate
- C) They always know exactly what they want
- D) They prefer written specifications only

33. What example is given in the lecture for using the Prototyping Model?

[MCQ - choose one]

- A) A payroll system with stable requirements
- B) A university mobile app, using a sample screen to get student feedback
- C) A missile guidance system
- D) A hospital radiation therapy machine

34. Which of the following are reasons to use the Prototyping Model, according to the lecture?

[Select all that apply]

- A) Helps understand what the user really wants
- B) Saves time by avoiding big mistakes
- C) Great for projects where requirements are unclear or may change
- D) Best for projects with completely fixed, well-known requirements

35. In the Prototyping Model, what allows the client to explain their needs more clearly?

[MCQ - choose one]

- A) Reading a written specification document
- B) Interacting with a working dummy version of the system
- C) Attending a lecture on software engineering
- D) Reviewing the final source code

36. What is a key risk when using the Prototyping Model, according to the model comparison table?

[MCQ - choose one]

- A) Requirements are always frozen
- B) The prototype may be mistaken as the final product, leading to scope creep
- C) It requires too much upfront planning
- D) It cannot involve the customer at all

37. In the Incremental Model, how is the system developed?

[MCQ - choose one]

- A) All at once in a single release
- B) Step by step in small parts called increments
- C) Only through risk analysis loops
- D) Exclusively through a prototype

38. In the Incremental Model, what does each increment go through?

[MCQ - choose one]

- A) Only the testing phase
- B) The entire SDLC (Requirements -> Design -> Implementation -> Testing -> Deployment)
- C) Only the design phase
- D) Only requirements gathering

39. What real-world example is used to illustrate the Incremental Model?

[MCQ - choose one]

- A) Therac-25 radiation machine
- B) Windows operating system (Windows 7 -> 8 -> 10 -> 11)
- C) A single payroll system release
- D) The Ariane 5 rocket

40. According to the lecture, the Incremental Model is suitable when:

[Select all that apply]

- A) Working on very large projects
- B) Most requirements are known, but the system needs to be developed in phases
- C) Quick delivery of working features and early user feedback is needed
- D) The project is extremely simple with only one small function

41. According to the lecture recording, why might a company use the Incremental Model for a broad, general-audience product (like an operating system)?

[MCQ - choose one]

- A) Because it targets one specific client with fixed needs
- B) Because it will be used by many different people, and functions can be released gradually
- C) Because it avoids all testing
- D) Because it requires no requirements gathering

42. In the Incremental Model, what can happen to earlier increments while later ones are being developed?

[MCQ - choose one]

- A) They cannot be changed at all
- B) They can be improved based on customer feedback
- C) They are deleted
- D) They are merged into the final release only

43. What does each 'increment' add to the system, according to the lecture?

[MCQ - choose one]

- A) A completely new, unrelated system
- B) New features on top of the previous increment
- C) Nothing; increments are identical
- D) Only bug fixes, never new features

44. What is the Spiral Model best described as?

[MCQ - choose one]

- A) A purely linear model
- B) A risk-focused and iterative approach combining Waterfall and Prototyping features
- C) A model with no testing at all
- D) A model used only for small, simple projects

45. What are the four main sections (quadrants) of each loop in the Spiral Model?

[Select all that apply]

- A) Planning
- B) Risk Analysis
- C) Engineering
- D) Evaluation
- E) Deployment only

46. What happens during the 'Risk Analysis' quadrant of a Spiral Model loop?

[MCQ - choose one]

- A) The system is deployed to all users
- B) Potential risks are identified and steps are taken to reduce or eliminate them
- C) The final code is written
- D) The client signs the contract

47. What happens during the 'Evaluation' quadrant of a Spiral Model loop?

[MCQ - choose one]

- A) The customer reviews the work done, and the next loop is planned based on feedback
- B) The requirements are frozen permanently
- C) The project is cancelled
- D) Only unit testing occurs

48. According to the lecture, the Spiral Model is recommended for:

[Select all that apply]

- A) Large, complex, or high-risk projects
- B) Projects where requirements are not fully known and may change
- C) Projects needing frequent reviews, testing, and updates
- D) Very small, low-risk, one-person projects

49. Which types of systems are given as examples where the Spiral Model is appropriate?

[Select all that apply]

- A) Defense systems
- B) Aerospace / NASA-related systems
- C) Missile guidance software
- D) Simple personal blogs

50. According to the lecture recording, why is the Spiral Model described as a 'combination of several models'?

[MCQ - choose one]

- A) Because it uses only Agile principles
- B) Because it incorporates SDLC, prototyping, and incremental delivery within its loops
- C) Because it has no relationship to any other model
- D) Because it replaces the need for SDLC entirely

51. What is a key characteristic of the Agile Model?

[MCQ - choose one]

- A) Rigid, one-time delivery with no customer involvement
- B) Iterative and incremental development with team collaboration, customer involvement, and flexibility to change
- C) A model that freezes requirements at the start
- D) A model used only for hardware development

52. What is a 'sprint' in the Agile Model?

[MCQ - choose one]

- A) A single long phase lasting an entire year
- B) A short, fixed-length cycle (usually 1-4 weeks) that delivers a working product
- C) A type of testing document
- D) A stakeholder meeting held once

53. Which of the following are the two most popular Agile frameworks mentioned in the lecture?

[Select all that apply]

- A) Scrum
- B) Kanban
- C) Waterfall

D) V-Model

54. What is the foundational document that Agile values and principles are based on?

[MCQ - choose one]

- A) The SWEBOK Guide
- B) The Agile Manifesto (2001)
- C) The ACM Code of Ethics
- D) IEEE Software Standard 730

55. According to the lecture, what makes Agile increasingly popular among companies today?

[MCQ - choose one]

- A) It avoids customer feedback entirely
- B) It allows customers to change requirements frequently and get quick releases
- C) It requires no trained team members
- D) It eliminates the need for testing

56. What is required for a development team to successfully follow Agile methodology, according to the lecture recording?

[MCQ - choose one]

- A) No special preparation is needed
- B) The team needs to be trained in Agile principles
- C) The team must use only the Waterfall model first
- D) The team must avoid all customer contact

57. Which of the following are described as key roles in the Scrum framework?

[Select all that apply]

- A) Product Owner - chooses what to build
- B) Scrum Master - guides the team
- C) Team - builds the product
- D) Business Analyst - sole owner of testing

58. Which model is described as 'Flow-based' with 'Continuous' delivery, best suited for maintenance and support workflows in the Model Comparison table?

[MCQ - choose one]

- A) Scrum (Agile)
- B) Kanban (Agile)
- C) Spiral
- D) V-Model

59. In the COVID-19 vaccination appointment system activity, which models were ruled out, and why (based on the lecture's reasoning)?

[Select all that apply]

- A) Waterfall Model - requires all requirements upfront, no early delivery, no flexible changes
- B) V-Model - too heavy on testing, not fast enough for an urgent rollout
- C) Prototyping Model - too time-consuming to build a prototype before starting real development
- D) Incremental Model - ruled out because it cannot handle changing requirements

60. According to the lecture's analysis, which model was identified as the best choice for the COVID-19 vaccination appointment system scenario, and why?

[MCQ - choose one]

- A) Spiral Model, because it focuses purely on risk
- B) Incremental Model, because it enables early delivery, handles changing requirements, and allows feedback-based updates
- C) Classical Waterfall Model, because requirements were fully known in advance
- D) V-Model, because heavy testing was the top priority

Answer Key

1. B
2. B
3. A, B, C
4. B
5. B
6. B
7. B
8. C
9. A, B, C
10. B
11. A, B, C
12. C
13. B
14. C
15. B
16. B
17. A, B, C
18. B
19. A
20. B
21. B
22. B
23. B
24. D
25. A, B, C
26. B
27. B
28. B
29. B
30. A, B, C
31. B
32. B
33. B

34. A, B, C

35. B

36. B

37. B

38. B

39. B

40. A, B, C

41. B

42. B

43. B

44. B

45. A, B, C, D

46. B

47. A

48. A, B, C

49. A, B, C

50. B

51. B

52. B

53. A, B

54. B

55. B

56. B

57. A, B, C

58. B

59. A, B, C

60. B